Synthetic Dataset Generation Strategy

AI-Driven Power Grid Stability Research

1. The Synthetic Data Rationale

In power systems research, real-world "Collapse" data is scarce because grid operators work hard to prevent it. Synthetic data generation allows you to create a **balanced dataset** that includes thousands of simulated failure scenarios, which is essential for training a robust Machine Learning model.

2. Sample Data Schema (CSV Structure)

Feature	Description	Unit	Normal Range
Frequency	System frequency	Hz	49.75 – 50.25
Voltage_Mag	Bus voltage magnitude	kV	313.5 – 346.5
P_Load	Total Active Power demand	MW	Variable
Q_Load	Total Reactive Power demand	MVAr	Variable
Stability_Label	Target Variable (0 or 1)	Binary	0 (Stable) / 1 (Unstable)

3. Simulation Workflow (IEEE 14-Bus System)

The IEEE 14-bus system serves as a "Digital Twin" for the grid. By modifying its parameters to simulate Nigerian conditions (high technical losses and radial topology), you can generate high-fidelity data.

Generation Loop:

- **Step 1:** Randomly vary load demands across all buses (Normal Distribution).
- **Step 2:** Introduce "Events" (e.g., disconnecting a transmission line or a generator).
- **Step 3:** Execute Power Flow analysis using a tool like *Pandapower* or *MATLAB*.
- **Step 4:** Check Frequency stability. If $f < 48.5\ Hz$, label as "1".

4. Implementation Snippet (Python Example)

```
import pandapower as pp
import numpy as np

# Create an IEEE 14-bus network
net = pp.networks.case14()

def generate_data(n_samples=1000):
    data = []
    for _ in range(n_samples):
        # Randomly scale loads between 80% and 120%
        scaling_factor = np.random.uniform(0.8, 1.2)
        net.load.p_mw *= scaling_factor

        try:
            pp.runpp(net) # Run Power Flow
            freq = 50.0 + np.random.normal(0, 0.05) # Simulated freq deviation
            stable = 1 if freq > 49.0 else 0
            data.append([freq, net.res_bus.vm_pu.mean(), stable])
        except:
            # Convergence failure often implies total instability
            data.append([0.0, 0.0, 1])
    return data
```

5. Novelty for Nigerian Context

Your research is novel because it uses **Physics-Informed Data Synthesis**. Instead of using generic public data, you are building a dataset that reflects the specific technical constraints of the Nigerian energy landscape, such as high ambient temperatures and localized radial bottlenecks.